جامعة بيرزيت

BIRZEIT UNIVERSITY

SP.TOP: DATA SCIENCE AND ANALYTICS
COMP4381
Project Third Phase
Flight Delay Prediction Using Machine Learning

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Section 2

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Abstract

Flight delays are a major challenge in the aviation industry, affecting passengers, airlines, and airport operations. In this project, flight delays will be studied in the context of European flights, using historical weather information collected from airport. This project aims to study whether weather conditions contribute significantly to flight delays, and to identify patterns associated with cities, seasons, and weather types. Python language and data analysis libraries were used for preprocessing data, feature engineering, statistical analysis, and visualization.

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1. Introduction

There are many reasons for flight delays, including weather conditions, air traffic control, technical malfunctions, or operational procedures Understanding the factors that influence delays contributes to improved flight preparedness at airports and airlines.

This project combines flight logs and weather information to study the relationship between weather conditions and flight delays. The analysis focuses on five major European cities and includes data collected between January 2023 and December 2025.

2. Dataset Description

The final dataset contains flight records enriched with weather information obtained from airport locations.

Dataset Characteristics

- Number of records: 163,651 flights
- Number of features: 55 columns
- Target variable: `delay_min`
- Date range: January 2023 – December 2025

Selected Cities

Table 1 Selected Cities Included in the Dataset

City	Country
Amsterdam	Netherlands
Barcelona	Spain
Frankfurt	Germany
London	United Kingdom
Paris	France

Number of Flights by City

Table 2 Number of Flights by Origin City

City	Flights
London	49,206
Amsterdam	33,191
Paris	30,837
Frankfurt	26,031
Barcelona	24,386

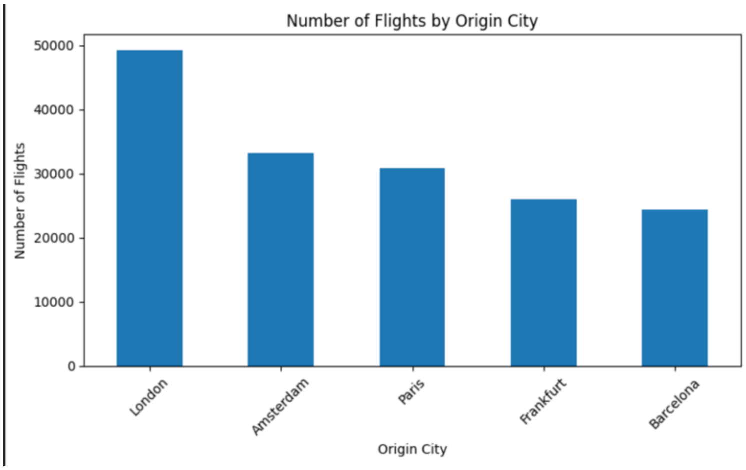

Figure 1 : Number of Flights by Origin City

3. Data Sources

This project combines three separate datasets to analyze flight delays. the primary dataset is based on the “Flight Delays Europe 2023–2025” dataset on Hugging Face platform. The dataset includes flight dates, departure and arrival airports, routes, scheduled and actual flight times, and specific delay indicators. These operational records serve as the foundation for the entire study. To provide geographical context, airport metadata was extracted from the OurAirports database. This metadata includes coordinates, city and country names, elevations, airport types, and scheduled service details.

Linking this metadata to flight logs enabled regional delay analysis and provided the precise locations needed to collect weather data. Daily weather logs were drawn from the Open-Meteo Historical Weather API. Using the latitude and longitude of each departure airport, along with the specific flight dates, the API provided daily data for temperature, precipitation, rain, snow, wind speed, gusts, and weather codes. Combining these three data sources creates a single dataset encompassing operations, geography, and weather, setting the stage for initial exploratory analysis and future forecasting modeling.

4. Data Preparation

Several pre-treatment steps were performed before the analysis:

Data Cleaning

These procedures were performed before analyzing the dataset to enhance data quality and obtain reliable results.

Duplicate data was removed from the database to prevent it from affecting the analysis. Then, missing values were checked for all variables in the dataset. Columns containing missing values or low analytical value were deleted because they lacked useful data.Metadata about airports was merged with the flights' dataset using airport identifiers, and observations without airport data were excluded because no data regarding cities, countries, or geographical coordinates existed for those records.

Missing data in categorical variables with a large number of missing values was replaced with an "unknown" category. This allowed the project to retain useful flight data despite inconsistencies in some variables. Date variables were converted to a date-time data type to enable the creation of time-specific properties. Data consistency for airport IDs, routes, weather variables, and dates was verified before property creation began.

Feature Engineering

Additional features were generated:

- Year
- Month
- Weekday
- Season
- Delay category
- Route frequency
- Traffic level
- Weather indicators
- Weather condition groups

Delay Categories

Flights were categorized as:

- Early or On Time
- Small Delay
- Moderate Delay
- Long Delay

5. Weather Data Integration

Weather information was collected using airport coordinates and merged with flight records.

The following weather features were added:

- Mean temperature
- Maximum temperature
- Minimum temperature
- Rainfall
- Snowfall
- Wind speed
- Wind gusts
- Weather code
- Weather category

Additional binary indicators were created:

- Rainy day
- Snowy day
- Windy day
- Heavy precipitation day

A total of 15 weather-related features were added to the dataset.

6. Exploratory Data Analysis

Delay Statistics

Table 3 Summary Statistics of Flight Delays

Metric	Value
Mean Delay	1.59 min
Median Delay	0.08 min
Minimum Delay	-120.00 min
Maximum Delay	824.50 min
Standard Deviation	12.37 min

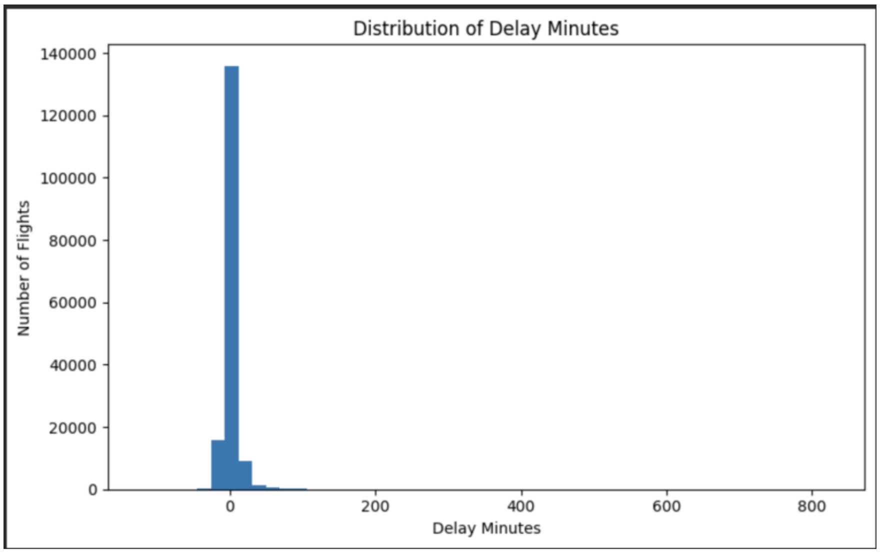

Figure 2 Distribution of Flight Delay Minutes

Interpretation

The average flight delay was only 1.59 minutes. The average delay was close to zero, indicating that most flights experienced very minimal delays. However, some extreme delays exceeded 800 minutes, increasing the overall variability.

Delayed Flights (>15 Minutes)

Table 4 Percentage of Delayed and Non-Delayed Flights

Category	Percentage
Not Delayed	95.17%
Delayed	4.83%

Interpretation

Only 4.83% of flights experienced delays exceeding 15 minutes, suggesting that significant delays were relatively rare.

Delay Categories Distribution

Table 5 Flight Counts by Delay Category

Delay Category	Number of Flights
Early or On Time	81,062
Small Delay	74,682
Moderate Delay	7,115
Long Delay	792

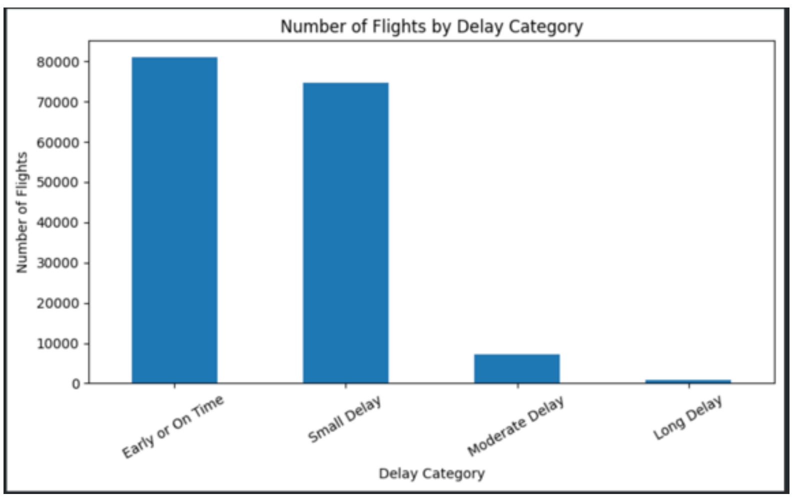

Figure 3 Number of Flights by Delay Category

Interpretation

Most flights were either on time or experienced only minor delays. Long delays accounted for less than 1% of all flights.

7. Weather Impact Analysis

To evaluate weather influence, weather conditions were grouped into several categories.

Average Delay by Weather Condition

Table 6 Delay Statistics by Weather Condition Group

Weather Condition	Flights	Average Delay	Delayed Rate (%)
Clear	4,263	1.93	5.63
Cloudy	62,321	1.87	5.21
Drizzle	61,339	1.52	4.68
Snow	4,012	1.35	5.03
Rain	31,716	1.16	4.24

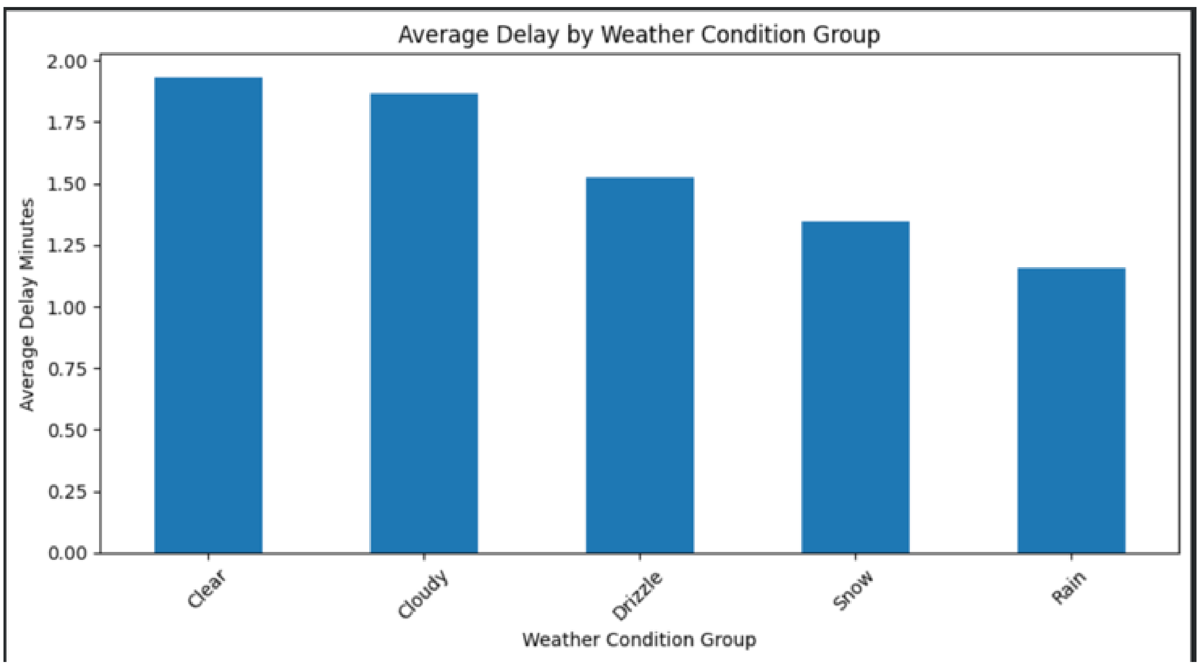

Figure 4 Average Delay by Weather Condition Group

Interpretation

Unexpectedly, clear weather showed the highest average delay, while rainy conditions recorded the lowest average delay. This indicates that weather alone does not explain delay behavior and that operational factors likely play an important role.

Bad Weather vs Normal Weather

Weather conditions were classified into:

- Normal Weather
- Bad Weather

The comparison showed that bad weather days did not consistently cause greater delays than normal weather days.

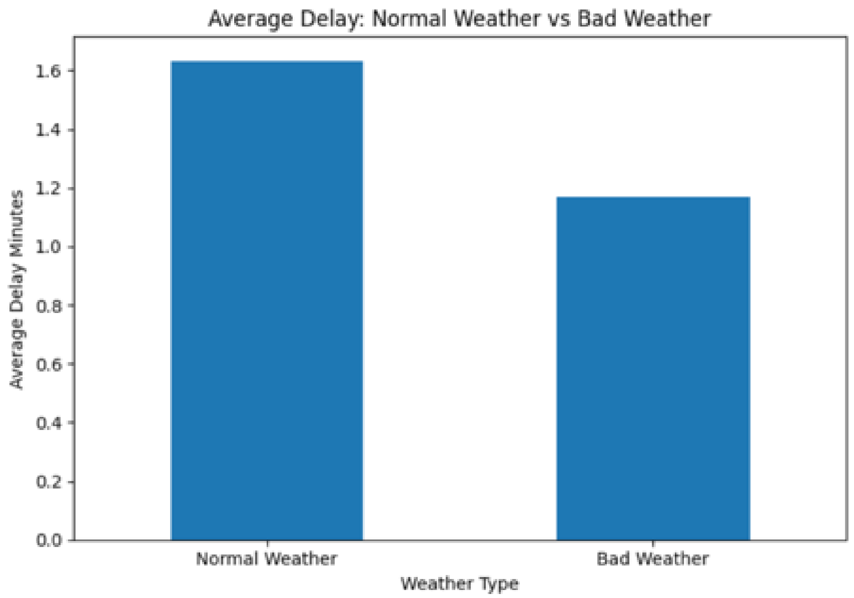

Figure 5 Average Delay: Normal Weather vs Bad Weather

Interpretation

The results suggest that airports and airlines may be able to adapt their operations during adverse weather conditions, reducing their impact on delays.

8. Seasonal Analysis

Weather effects were also examined across seasons.

Table 7 Average Delay by Season and Weather Type

Season	Weather Type	Average Delay
Autumn	Bad Weather	1.13
Autumn	Normal Weather	1.63
Spring	Bad Weather	1.41
Spring	Normal Weather	2.03
Summer	Bad Weather	1.27
Summer	Normal Weather	1.17
Winter	Bad Weather	0.91
Winter	Normal Weather	1.77

Interpretation

Seasonal variation was observed during the analysis of average delays between adverse and normal weather conditions. The results indicate that the relationship between weather and delays differs according to the season.

During the summer, the average delay for flights operating in adverse weather was slightly higher than for those operating in good weather. However, in autumn, spring, and winter, the opposite was true, with average flight delays on adverse weather being lower.

From these findings, we understand that weather is not the sole cause of flight delays. Delays are influenced by numerous other factors, such as seasonal demand, airport congestion, route structure, and weather. Weather is just one of many variables that affect flight delays.

9. Correlation Analysis

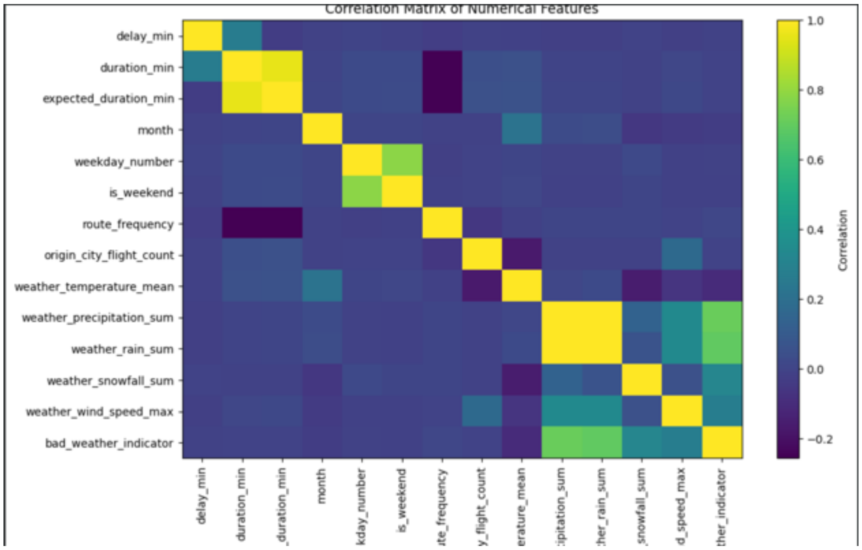

Figure 6 Correlation Matrix of Numerical Features

Interpretation

Correlation matrices were created to examine any relationships between numerical data attributes within the dataset. Many variables showed weak to moderate correlations with the target variable, suggesting that multiple variables work together to influence flight delays, rather than one dominant factor.

Some relationships were observed between weather-related variables, including precipitation, rain, snow, and wind speed, and delay behavior; however, none of these relationships were significant enough to independently explain the delay pattern. Operational and time-related variables also influenced the observed variability.

This analysis further confirms that flight delays are a complex process influenced by weather, airport traffic, routes, and operational factors operating simultaneously.

10. Aggregation Effect Discussion

One of the unexpected findings was the relationship between weather conditions and flight delays. Conventional aviation research suggests that bad weather naturally increases delays. However, the data compiled in this project showed that days with bad weather did not consistently have higher average delays than days with fair weather. While this may seem illogical, this discrepancy points to the clustering effect, as in Simpson's paradox. This paradox occurs when a trend observed in a large dataset changes or disappears entirely when that data is divided into smaller groups.

To test this, the data were segmented by specific cities and seasons. The results showed a mix of patterns. In certain cities and during specific seasons, bad weather was directly associated with higher delays and longer waiting times. In other cities, the opposite was true. Combining all these separate locations into a single average masked these local variations, distorting the overall picture.

This variability demonstrates why flight delays need to be analyzed within specific contexts rather than relying on general averages. It underscores that delays are caused by a combination of factors—such as runway traffic, route types, peak seasonal travel, and airline operations—that work in conjunction with weather conditions, rather than weather being the sole cause.

11. Findings

The main findings of this project are:

1. Most flights were on time or experienced only minor delays.
2. Only 4.83% of flights were delayed by more than 15 minutes.
3. Weather conditions showed limited impact on average delay times.
4. Clear weather sometimes exhibited higher average delays than adverse weather.
5. Seasonal variations existed but were not strong enough to establish weather as the dominant factor.
6. Other operational variables likely contribute more significantly to delay occurrence.
7. City-level and seasonal analyses revealed mixed weather-delay patterns, suggesting that aggregated weather statistics may hide local effects.

12. Limitations

This dataset covers a wide area encompassing more than 163,000 flight records, but you must consider five specific limitations when looking at these results.

First, weather records rely on daily summaries rather than hourly observations. Flight delays often depend on precise weather conditions at the time of departure, which a single daily average cannot capture.

Second, the data is limited to weather observations at departure airports only, completely ignoring weather conditions at arrival airports or along the actual flight path. This means that these atmospheric influences are entirely absent from this analysis.

Third, data on many operational variables is lacking. The dataset does not account for runway congestion, unscheduled aircraft maintenance, staffing shortages, air traffic control restrictions, gate congestion, or individual airline decisions. Any of these factors can cause delays on their own, regardless of weather conditions.

Fourth, this phase of the project is limited to exploratory and descriptive analysis. These patterns show correlations, not direct causes, so establishing conclusive causal relationships requires further statistical testing and predictive modeling.

Finally, despite the large sample size, this data remains a specific part of Eurocontrol's complete records and may miss some broader flight delay patterns.

13. Conclusion

This project analyzed more than 163,000 flight records supported by information from five major European cities. Although weather conditions affect flight operations, the results indicate that weather alone does not fully explain flight delays. Large delays were relatively rare, and bad weather conditions did not consistently cause delays longer than normal weather conditions.

Future work may include additional variables such as airport congestion, airline flight schedules, aircraft type, and air traffic management data to improve predictive performance and better understand the causes of flight delays.

References

Hugging Face. Flight Delays Europe 2023–2025 Dataset. Available at: **https://huggingface.co/datasets/345rf4gt56t4r3e3/flight-delays-europe-2023-2025**

OurAirports. Airports Database. Available at: **https://ourairports.com/data/**

Open-Meteo. Historical Weather API Documentation. Available at: **https://open-meteo.com/**

Pandas Development Team. Pandas Documentation. Available at: **https://pandas.pydata.org/**

NumPy Developers. NumPy Documentation. Available at: **https://numpy.org/**

Matplotlib Development Team. Matplotlib Documentation. Available at: **https://matplotlib.org/**